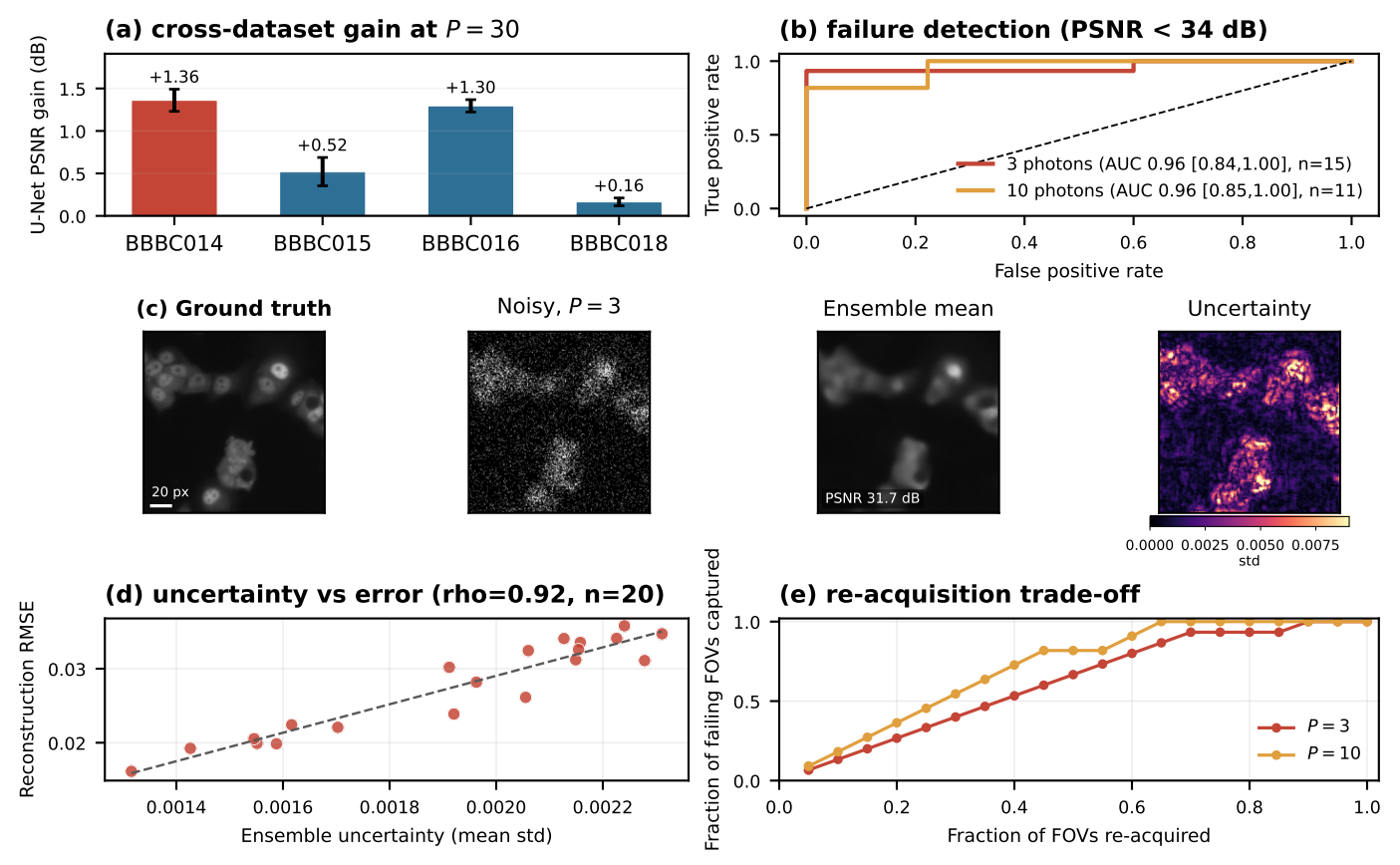

(a) cross-dataset gain at $P = 30$
U-Net PSNR gain (dB)
+1.36
+0.52
+1.30
+0.16
BBBC014
BBBC015
BBBC016
BBBC018
(b) failure detection (PSNR < 34 dB)
True positive rate
False positive rate
3 photons (AUC 0.96 [0.84,1.00], n=15)
10 photons (AUC 0.96 [0.85,1.00], n=11)
(c) Ground truth
20 px
Noisy, $P = 3$
Ensemble mean
PSNR 31.7 dB
Uncertainty
0.0000 0.0025 0.0050 0.0075
std
(d) uncertainty vs error (rho=0.92, n=20)
Reconstruction RMSE
Ensemble uncertainty (mean std)
(e) re-acquisition trade-off
Fraction of failing FOVs captured
Fraction of FOVs re-acquired
$P = 3$
$P = 10$